Lab 10: Gradient Descent and Convexity Solutions

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, November 11th, 2026

Name: _____

uniqname: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section.

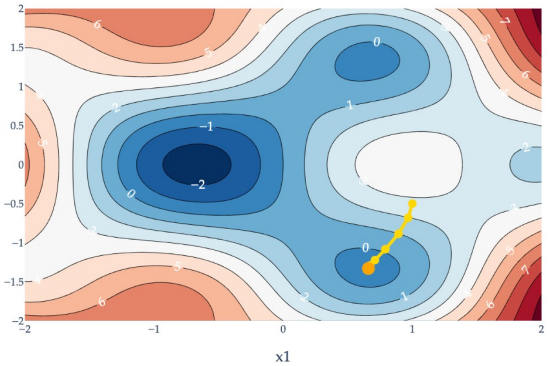
While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Recap: Gradient Descent (Chapter 8.3)

Suppose $f : \mathbb{R}^d \to \mathbb{R}$ is a **differentiable** vector-to-scalar function, meaning that all of its partial derivatives are defined everywhere. To find $\vec{x}^*$, the **minimizer** of f:

1. Choose a positive number, α. This number is called the **learning rate**, or **step size**.
2. Choose an **initial guess** for the minimizer, $\vec{x}^{(0)}$.
3. Then, repeatedly update the guess using the **update rule**:
$$\vec{x}^{(t+1)} = \vec{x}^{(t)} - \alpha \nabla f(\vec{x}^{(t)})$$
4. Terminate once the algorithm converges, which happens when the norm of the gradient, $\|\nabla f(\vec{x}^{(t)})\|$, is below some small **tolerance** level, e.g. 0.001 (since this must mean we're very close to a minimum).

Intuition: The gradient vector $\nabla f(\vec{x}^{(t)})$ tells us the direction of greatest increase in f at the current guess $\vec{x}^{(t)}$, so $-\nabla f(\vec{x}^{(t)})$ is the direction that will **decrease** our function the most. The distance moved in that direction is determined by the step size α, which scales $-\nabla f(\vec{x}^{(t)})$. To update our guess, we add $-\alpha \nabla f(\vec{x}^{(t)})$ to the old guess, $\vec{x}^{(t)}$. Then, we repeat this process.



Example gradient descent path for a function $f : \mathbb{R}^2 \to \mathbb{R}$.

Activity 1: Gradient Descent Gone Wrong

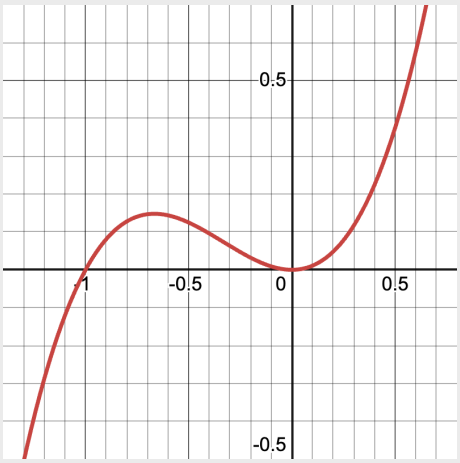
Suppose $\vec{x} \in \mathbb{R}^2$. Let

$$f(\vec{x}) = x_1^3 + \|\vec{x}\|^2 = x_1^3 + x_1^2 + x_2^2$$

To minimize $f(\vec{x})$, we use gradient descent, with a learning rate of $\alpha = \frac{1}{4}$.

a) Open Desmos and plot the related function $g(x) = x^3 + x^2$. Even though this is a scalar-to-scalar function, and f is vector-to-scalar, they are related. What do you notice about the shape of the graph?

Solution: $g(x)$ is a cubic function, with a local minimum and local maximum and no global minimum nor maximum. It is not convex.



b) Find $\nabla f(\vec{x})$, the gradient of $f(\vec{x})$.

Solution:

$$\nabla f(\vec{x}) = \begin{bmatrix} 3x_1^2 + 2x_1 \\ 2x_2 \end{bmatrix} = \begin{bmatrix} 3x_1^2 \\ 0 \end{bmatrix} + 2\vec{x}$$

c) Recall, $\vec{x}^{(t)}$ is the guess for $\vec{x}^*$ at timestep t. Let $\vec{x}^{(t)} = \begin{bmatrix} x_1^{(t)} \\ x_2^{(t)} \end{bmatrix}$.

Show that

$$x_1^{(t+1)} = \frac{1}{2}x_1^{(t)} - \frac{3}{4}(x_1^{(t)})^2, \qquad x_2^{(t+1)} = \frac{1}{2}x_2^{(t)}$$

Solution: The gradient descent update rule states that $\vec{x}^{(t+1)} = \vec{x}^{(t)} - \alpha \nabla f(\vec{x}^{(t)})$. But, since $\nabla f(\vec{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}$, we can think of the update rule as being two separate update rules:

$$x_i^{(t+1)} = x_i^{(t)} - \alpha \frac{\partial f}{\partial x_i}(\vec{x}^{(t)}), \quad i = 1, 2$$

Then,

$$\begin{aligned} x_1^{(t+1)} &= x_1^{(t)} - \alpha \frac{\partial f}{\partial x_1}(\vec{x}^{(t)}) \\ &= x_1^{(t)} - \frac{1}{4}(3(x_1^{(t)})^2 + 2x_1) \\ &= x_1^{(t)} - \frac{3}{4}(x_1^{(t)})^2 - \frac{2}{4}x_1^{(t)} \\ &= \frac{1}{2}x_1^{(t)} - \frac{3}{4}(x_1^{(t)})^2 \end{aligned}$$

$$\begin{aligned} x_2^{(t+1)} &= x_2^{(t)} - \alpha \frac{\partial f}{\partial x_2}(\vec{x}^{(t)}) \\ &= x_2^{(t)} - \frac{1}{4}(2x_2^{(t)}) \\ &= x_2^{(t)} - \frac{1}{2}x_2^{(t)} \\ &= \frac{1}{2}x_2^{(t)} \end{aligned}$$

d) For any initial guess $\vec{x}^{(0)}$, what does $x_2^{(t)}$ converge to as $t \to \infty$?

Solution: $x_2^{(t)}$ converges to 0 as $t \to \infty$ because it's being divided in half at each iteration.

e) Suppose $\vec{x}^{(0)} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$.

i) Find $\vec{x}^{(1)}$.

ii) Will gradient descent eventually converge, given this initial guess and learning rate?

Solution:
i. Recall that $x_2^{(t+1)} = \frac{1}{2}x_2^{(t)}$. So, $x_2^{(1)} = \frac{1}{2}x_2^{(0)} = \frac{1}{2} \cdot 0 = 0$. As for $x_1^{(1)}$, we have

$$\begin{aligned} x_1^{(1)} &= \frac{1}{2}x_1^{(0)} - \frac{3}{4}(x_1^{(0)})^2 \\ &= \frac{1}{2}(-1) - \frac{3}{4}(-1)^2 \\ &= -\frac{1}{2} - \frac{3}{4} \\ &= -\frac{5}{4} \end{aligned}$$

ii. Gradient descent will not converge: the guesses for $x_1^{(t)}$ will get larger and larger in magnitude, approaching $-\infty$. This happens because the term $-\frac{3}{4}(x_1^{(t)})^2$ dominates the term $\frac{1}{2}x_1^{(t)}$ in the iteration rule. Since $x_1^{(1)}$ is already larger than 1 in magnitude, $(x_1^{(1)})^2$ will be even larger than that (since when you square a number greater than 1, it increases in size), which propagates to the next iteration and so on.

$$x_1^{(t+1)} = \frac{1}{2}x_1^{(t)} - \frac{3}{4}(x_1^{(t)})^2$$

f) Suppose $\vec{x}^{(0)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

i) Find $\vec{x}^{(1)}$.

ii) Will gradient descent eventually converge, given this initial guess and learning rate?

Solution: i. Similar to part **e)**, we only have to find $x_1^{(1)}$.

$$\begin{aligned} x_1^{(1)} &= \frac{1}{2}x_1^{(0)} - \frac{3}{4}(x_1^{(0)})^2 \\ &= \frac{1}{2}(1) - \frac{3}{4}(1)^2 \\ &= -\frac{1}{4} \end{aligned}$$

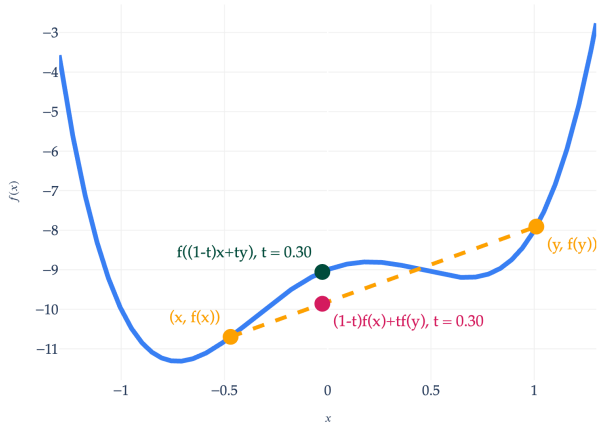
ii. Here, gradient descent **will** converge, since the absolute value of $x_1^{(1)}$ is less than 1. If we run more iterations, we'll see that $x_1^{(t)}$ is approacing zero because the absolute value keeps decreasing. $x_1^{(2)}$, for instance, is $-\frac{11}{64}$, and $|-\frac{11}{64}| < |-\frac{1}{4}|$.

Activity 2: Convexity and Gradient Descent

A function $f : \mathbb{R}^d \to \mathbb{R}$ is convex if for all $\vec{x}$ and $\vec{y}$ in its domain, and for any $t \in [0, 1]$,

$$f((1-t)\vec{x} + t\vec{y}) \leq (1-t)f(\vec{x}) + tf(\vec{y})$$

The English interpretation of this definition is that **the line connecting any two points on the graph of f always lies on or above the graph of f**. Intuitively, a convex function is a function that curves upward, like a bowl.



A non-convex function.

a) If a function is convex, it **must** have a global minimum.

◯ True ● False

> **Solution:** This is false. For example, $f(x) = e^x$ is convex, but it doesn't have a global minimum. It approaches 0 as $x \to -\infty$, but it never actually reaches it. $f(x) = x$ is also convex and doesn't have a global minimum.

b) If a function is convex, then gradient descent **must** converge on it given any initial guess and step size.

◯ True ● False

> **Solution:** This is also false. If we choose a step size α that is too large, gradient descent may oscillate or diverge, as we've seen in examples in Chapter 8.3.

c) If a function is convex, then any local minimum is also a global minimum.

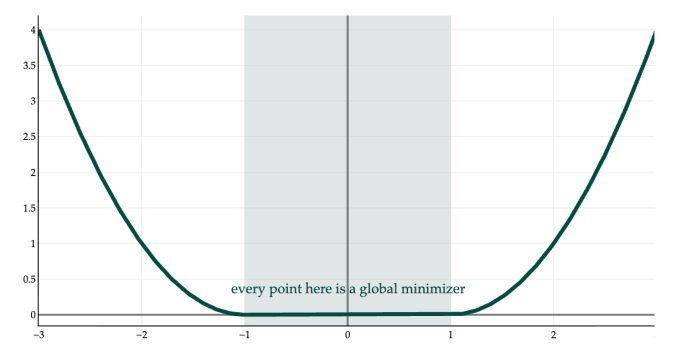
● True ◯ False

> **Solution:** This is true; a proof is given in Chapter 8.5.

d) If a function is convex and has a global minimum, then its minimizer **must** be unique.

◯ True ● False

Solution: This is false: a function can have multiple local minima that are all adjacent to each other, like a flat "ridge" at the very bottom.



Activity 3: Using Convexity to Prove Inequalities

Suppose $f : \mathbb{R} \to \mathbb{R}$ is a convex function.

a) Prove that $2f(5) \leq f(3) + f(7)$.

Solution: We'll start by solving for t using $x = 3$ and $y = 7$ with the expression for the input on the left side of the inequality.

$$\begin{aligned} 5 &= (1-t)x + ty \\ &= (1-t) \cdot 3 + 7t \\ &= 3 - 3t + 7t \\ &= 3 + 4t, \;\; t = \frac{1}{2} \end{aligned}$$

Then, plug the variables into the inequality and simplify.

$$\begin{aligned} f((1-\frac{1}{2}) \cdot 3 + \frac{1}{2} \cdot 7) &\leq (1-\frac{1}{2}) \cdot f(3) + \frac{1}{2} \cdot f(7) \\ f(\frac{3}{2} + \frac{7}{2}) &\leq \frac{f(3)}{2} + \frac{f(7)}{2} \\ f(5) &\leq \frac{f(3)}{2} + \frac{f(7)}{2} \\ 2f(5) &\leq f(3) + f(7) \end{aligned}$$

b) Suppose also that $f(0) = 0$. Prove that for all $y \in \mathbb{R}$ and $t \in [0, 1]$,

$$f(ty) \leq tf(y)$$

Solution: Since the definition of convexity states that

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$$

for all $x, y \in \mathbb{R}$, we can substitute whatever we'd like for x or y. Let's plug in $x = 0$. Then,

$$\begin{aligned} f((1-t) \cdot 0 + ty) &\leq (1-t)f(0) + tf(y) \\ f(ty) &\leq tf(y) \end{aligned}$$

Activity 4: Linear Approximation

Suppose $f : \mathbb{R} \to \mathbb{R}$, i.e. f is a scalar-to-scalar function. In general, the **tangent line** to $f(x)$ at $x = a$ is given by the equation

$$f(x) \approx \underbrace{f(a) + \left(\frac{\mathrm{d}f}{\mathrm{d}x}(a)\right)(x-a)}_{\text{tangent line}}$$

The $\approx$ symbol means that the tangent line is an approximation of $f(x)$ near $x = a$; in Appendix 2, we defined it as the **best linear approximation** of $f(x)$ near $x = a$. The expression on the right is a line with slope $\frac{\mathrm{d}f}{\mathrm{d}x}(a)$ that passes through the point $(a, f(a))$.

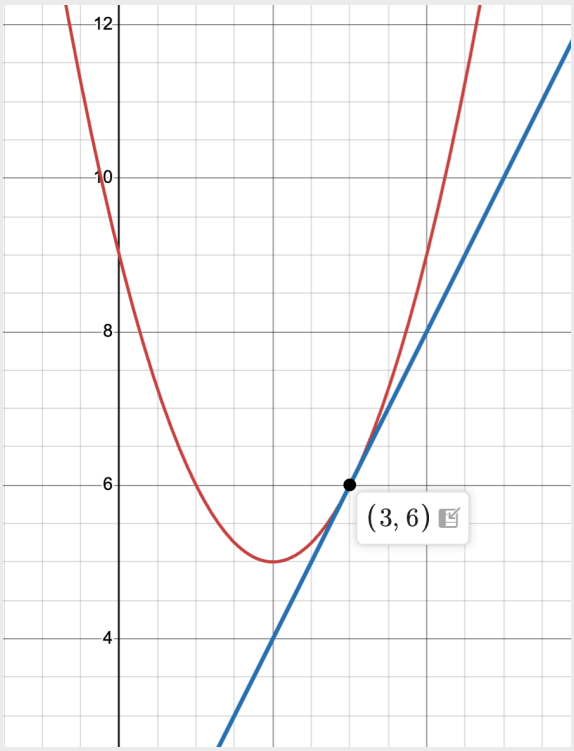
a) Draw $f(x) = (x-2)^2 + 5$ and its tangent line at $a = 3$.

Solution:
First, let's find the derivative of $f(x) = (x-2)^2 + 5$.

$$\frac{\mathrm{d}f}{\mathrm{d}x} = 2(x-2)$$

So, the tangent line to $f(x)$ at $a = 3$ is given by

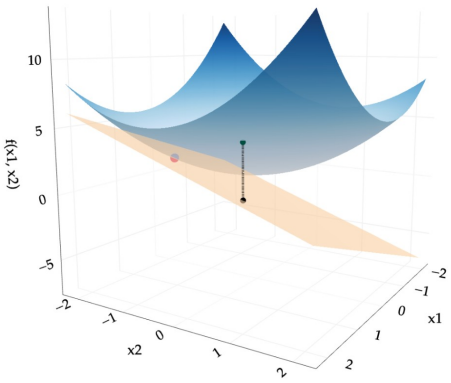
$$f(x) \approx f(3) + \left(\frac{\mathrm{d}f}{\mathrm{d}x}(3)\right)(x-3) = 6 + 2(3-2)(x-3) = 6 + 2x - 6 = \boxed{2x}$$



b) For vector-to-scalar functions, the best linear approximation at $\vec{a}$ is given by

$$L_{\vec{a}}(\vec{x}) = f(\vec{a}) + \nabla f(\vec{a})^T(\vec{x} - \vec{a})$$

If $\vec{x} \in \mathbb{R}^2$, this is called the **tangent plane**; if $\vec{x} \in \mathbb{R}^3$ or higher, it is called the **tangent hyperplane**.



Tangent plane for a function $f : \mathbb{R}^2 \to \mathbb{R}$.

Find the tangent plane to $f(\vec{x}) = \|\vec{x}\|^2 - 3$ at $\vec{a} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

Solution:
Let's start by finding the gradient of $f(\vec{x}) = \|\vec{x}\|^2 - 3$. Remember that $\nabla(\|\vec{x}\|^2) = 2\vec{x}$. So,

$$\nabla f(\vec{x}) = 2\vec{x}$$

as well.
So, the tangent plane to $f(\vec{x})$ at $\vec{a} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ is given by

$$\begin{aligned} L_{\vec{a}}(\vec{x}) &= f\left(\begin{bmatrix} 2 \\ 5 \end{bmatrix}\right) + \nabla f\left(\begin{bmatrix} 2 \\ 5 \end{bmatrix}\right)^T \left(\vec{x} - \begin{bmatrix} 2 \\ 5 \end{bmatrix}\right) \\ &= (2^2 + 5^2 - 3) + \left(2\begin{bmatrix} 2 \\ 5 \end{bmatrix}\right)^T \left(\vec{x} - \begin{bmatrix} 2 \\ 5 \end{bmatrix}\right) \\ &= 26 + \begin{bmatrix} 4 \\ 10 \end{bmatrix}^T \vec{x} - \begin{bmatrix} 4 \\ 10 \end{bmatrix}^T \begin{bmatrix} 2 \\ 5 \end{bmatrix} \\ &= 26 - (8 + 50) + \begin{bmatrix} 4 \\ 10 \end{bmatrix}^T \vec{x} \\ &= \boxed{\begin{bmatrix} 4 \\ 10 \end{bmatrix}^T \vec{x} - 32} \\ &= \boxed{4x_1 + 10x_2 - 32} \end{aligned}$$

The last two boxed expressions are both equivalent ways of writing the tangent plane. This plane passes through $(2, 5, f(2, 5))$ while having the same gradient as $f(\vec{x})$ at that point.

- Plugging in $x_1 = 2$ and $x_2 = 5$ into the final expression gives 26, which also matches the value of $f\left(\begin{bmatrix} 2 \\ 5 \end{bmatrix}\right) = 26$.
- In the latter form, the coefficients on x_1 and x_2 are 4 and 10, respectively, which match the components of the gradient vector $\nabla f\left(\begin{bmatrix} 2 \\ 5 \end{bmatrix}\right) = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$.

So, the boxed plane is **the** tangent plane to $f(\vec{x})$ at $\vec{a} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$. It's the best linear approximation of $f(\vec{x})$ near that point.

Activity 5: Gradient Descent for Empirical Risk Minimization

Suppose we have a dataset of 3 points, $(1, 2), (3, 5), (6, -1)$. We'd like to fit a simple linear regression model, $h(x_i) = w_0 + w_1 x_i$, to this dataset by minimizing average squared loss.

While we already know a closed-form solution for the optimal parameters — we've seen multiple equivalent versions of these formulas throughout the semester — let's use gradient descent to find them.

Let $\vec{w} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$. Using an initial guess of $\vec{w}^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and a step size of $\alpha = 0.1$, perform one iteration of gradient descent. What is $\vec{w}^{(1)}$?

Hint: You can proceed either by using the gradient of $R_{sq}(\vec{w}) = \frac{1}{n}\|\vec{y} - X\vec{w}\|^2$ from the notes or by computing the partial derivatives of $R_{sq}(w_0, w_1) = \frac{1}{n}\sum_{i=1}^{n}(y_i - (w_0 + w_1 x_i))^2$ with respect to w_0 and w_1.

Solution:
Let's use the gradient of $R_{\text{sq}}(\vec{w}) = \frac{1}{n}\|\vec{y} - X\vec{w}\|^2$ from the notes.

$$\nabla R_{\text{sq}}(\vec{w}) = -\frac{2}{n}(X^T\vec{y} - X^TX\vec{w}) = \frac{2}{n}X^T(X\vec{w} - \vec{y})$$

Here, $X = \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 1 & 6 \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$.
So, the gradient descent update is given by

$$\vec{w}^{(1)} = \vec{w}^{(0)} - \alpha\nabla R_{\text{sq}}(\vec{w}^{(0)}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \cdot \frac{2}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 6 \end{bmatrix}\left(\begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 1 & 6 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}\right) = \boxed{\begin{bmatrix} 2/5 \\ 11/15 \end{bmatrix}}$$

Note that if you did this using the partial derivatives of $R_{\text{sq}}(w_0, w_1) = \frac{1}{n}\sum_{i=1}^n (y_i - (w_0 + w_1x_i))^2$ with respect to w_0 and w_1, you would get the same answer!

$$\nabla R_{\text{sq}}(\vec{w}) = \begin{bmatrix} \frac{\partial R_{\text{sq}}}{\partial w_0} \\ \frac{\partial R_{\text{sq}}}{\partial w_1} \end{bmatrix} = \begin{bmatrix} -\frac{2}{n}\sum_{i=1}^n (y_i - (w_0 + w_1x_i)) \\ -\frac{2}{n}\sum_{i=1}^n (y_i - (w_0 + w_1x_i))x_i \end{bmatrix}$$

Plugging in $n = 3$, $x_1 = 1, x_2 = 3, x_3 = 6$, $y_1 = 2, y_2 = 5, y_3 = -1$, and $w_0^{(0)} = 0, w_1^{(0)} = 0$, we get

$$\begin{aligned}
\nabla R_{\text{sq}}(\vec{w}^{(0)}) &= \begin{bmatrix} -\frac{2}{n}\sum_{i=1}^n \left(y_i - (w_0^{(0)} + w_1^{(0)}x_i)\right) \\ -\frac{2}{n}\sum_{i=1}^n \left(y_i - (w_0^{(0)} + w_1^{(0)}x_i)\right)x_i \end{bmatrix} \\
&= \begin{bmatrix} -\frac{2}{3}\sum_{i=1}^3 (y_i - (0 + 0 \cdot x_i)) \\ -\frac{2}{3}\sum_{i=1}^3 (y_i - (0 + 0 \cdot x_i))\, x_i \end{bmatrix} \\
&= \begin{bmatrix} -\frac{2}{3}\sum_{i=1}^3 y_i \\ -\frac{2}{3}\sum_{i=1}^3 y_ix_i \end{bmatrix} \\
&= \begin{bmatrix} -\frac{2}{3}(2 + 5 + (-1)) \\ -\frac{2}{3}(2 \cdot 1 + 5 \cdot 3 + (-1) \cdot 6) \end{bmatrix} \\
&= \begin{bmatrix} -4 \\ -22/3 \end{bmatrix}
\end{aligned}$$

Then, $\vec{w}^{(1)}$ is given by

$$\begin{aligned}
\vec{w}^{(1)} &= \vec{w}^{(0)} - \alpha\nabla R_{\text{sq}}(\vec{w}^{(0)}) \\
&= \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1\begin{bmatrix} -4 \\ -22/3 \end{bmatrix} \\
&= \begin{bmatrix} 2/5 \\ 11/15 \end{bmatrix}
\end{aligned}$$

which is the same result as before!